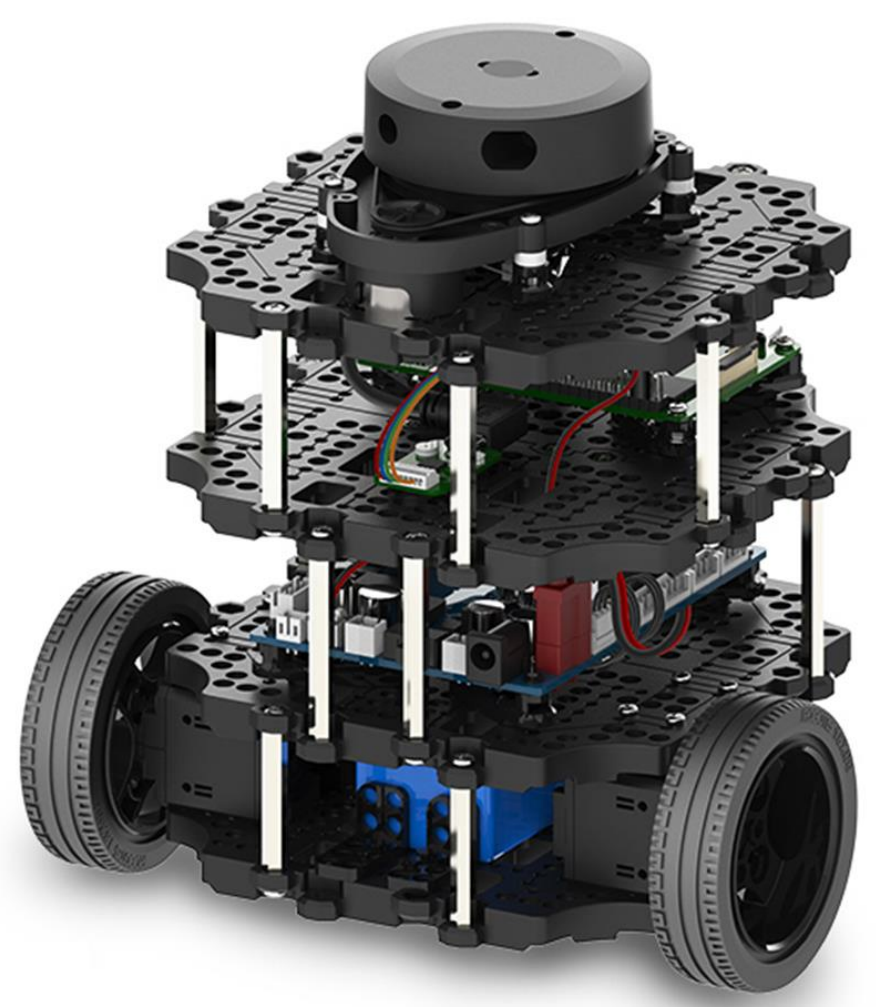


A Comparative Study of SLAM Algorithms for Robot Operating System

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MOTIVATION

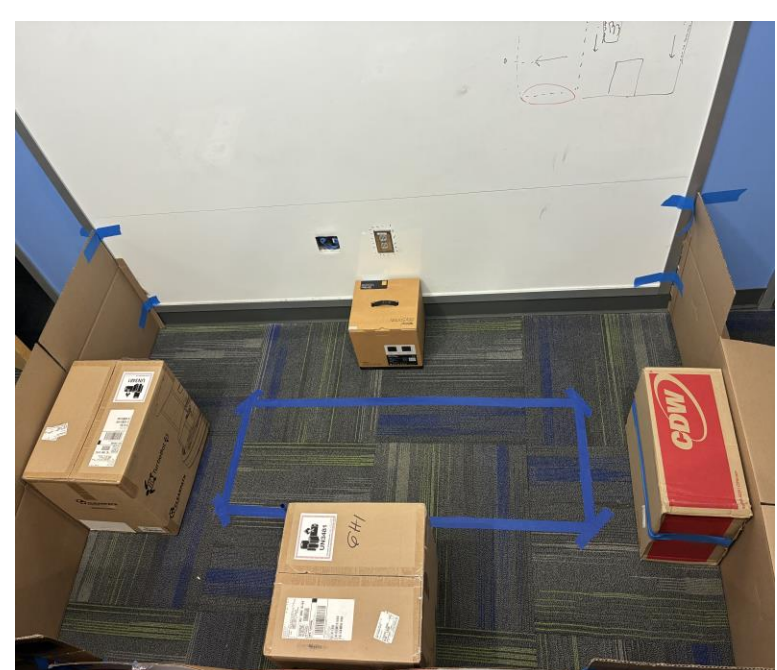
- Simultaneous Localization and Mapping (SLAM) is a useful algorithm for robot navigation and mapping of the environment at the same time
- Improving SLAM precision supports advanced applications such as autonomous exploration and loop-closure correction in dynamic environments
- Two popular SLAM algorithms, SLAM Toolbox [2] and Cartographer [3] produce different quality map under identical conditions, affecting localization and navigation

BACKGROUND/PRIOR WORKS

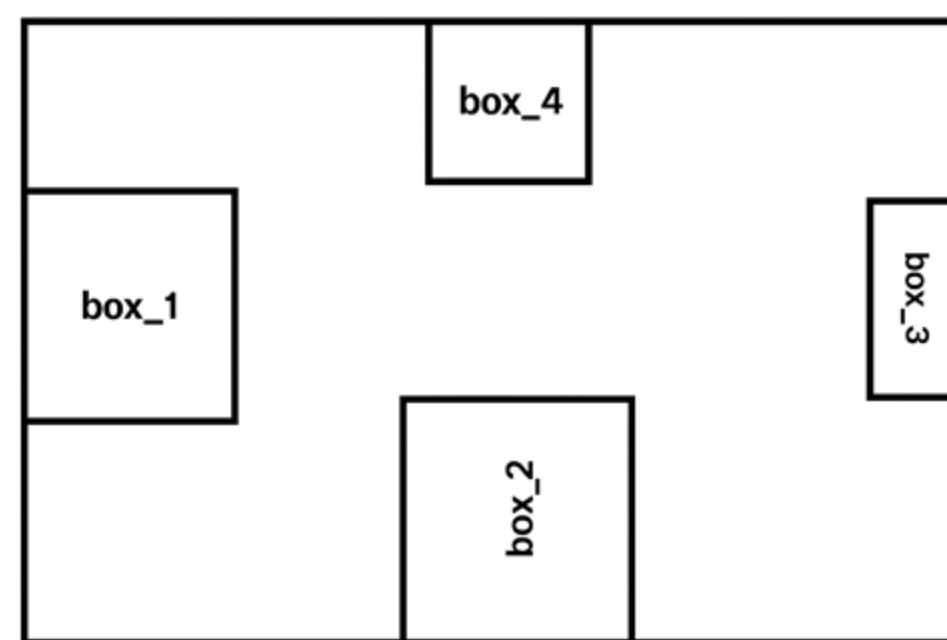
- Li & Zhu (2024) evaluated three 2D LiDAR SLAM algorithms Hector, GMapping, and Cartographer in simulated orchard environments using Gazebo
 - Cartographer achieved the best localization and mapping accuracy (≈ 8 cm error at 4 cm resolution), but required more CPU and memory
 - Performance metrics: Localization error RMSE, Mapping error MAE, CPU and memory usage

PROBLEM FORMULATION

- This study compares Cartographer and SLAM Toolbox to determine which SLAM algorithm generates more accurate and consistent maps under identical conditions
- Although both are widely used for autonomous navigation and mapping, performance differences arise from sensor noise, environment geometry, and algorithm design
- A controlled maze environment with fixed ground-truth dimensions was used to ensure fair testing and repeatable conditions



Test Environment

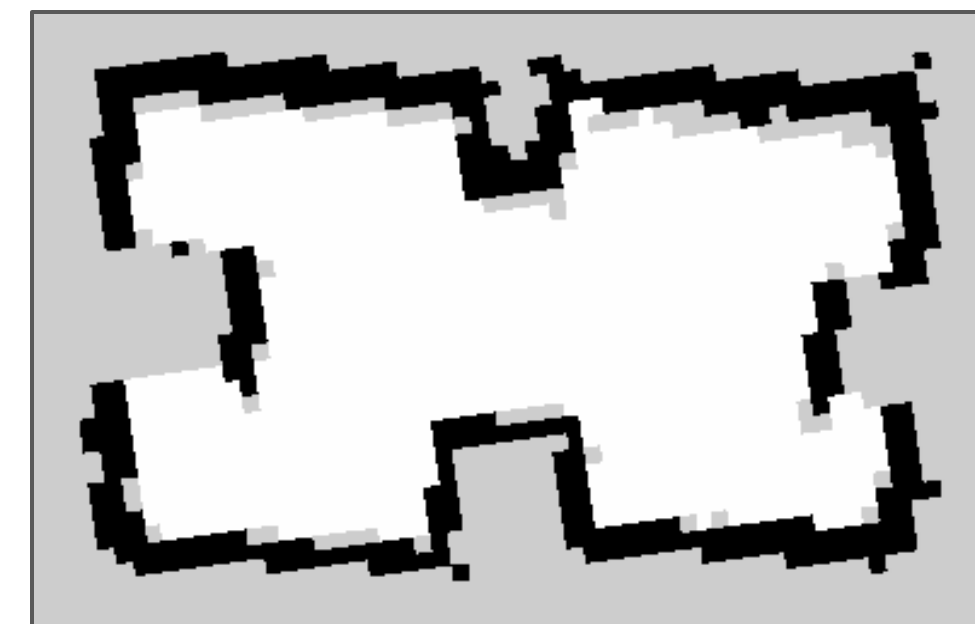


Model Environment

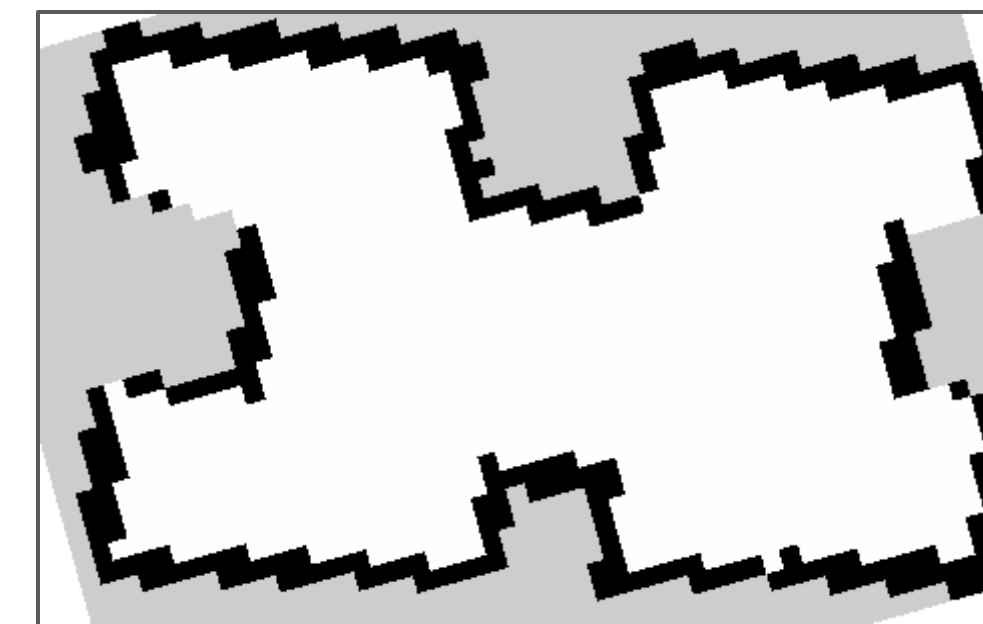
- The experiment quantifies mapping accuracy, reliability, and structural fidelity, identifying which algorithm provides superior geometric precision for the TurtleBot 3 platform

METHODOLOGY

- Both algorithms were configured using the default software setup from the original authors [1, 2]
- Using each algorithm, the real TurtleBot was driven over predefined path at the speed of 0.01 m/s
- Tests were performed 10 times in the same maze environment to ensure repeatability and gauge variance for each algorithm

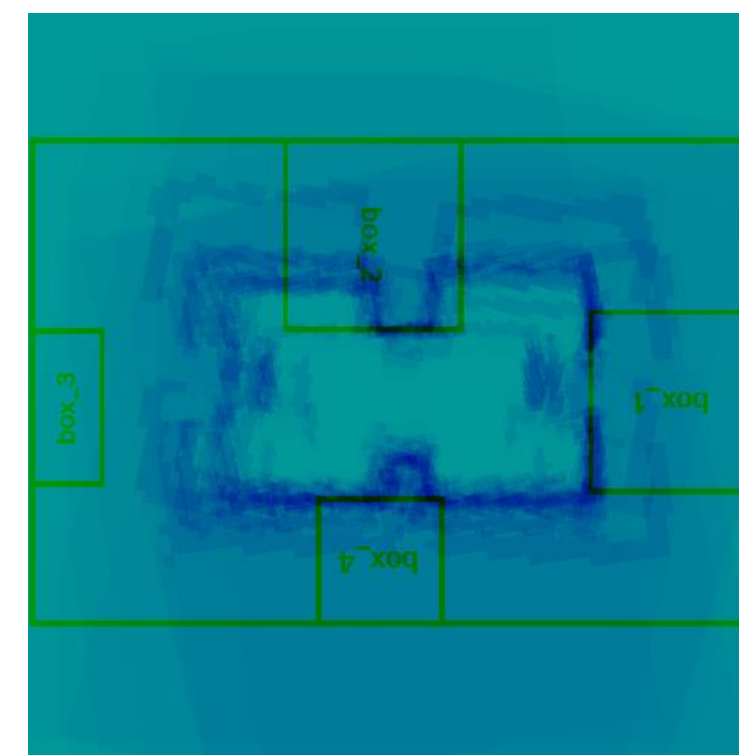


Cartographer

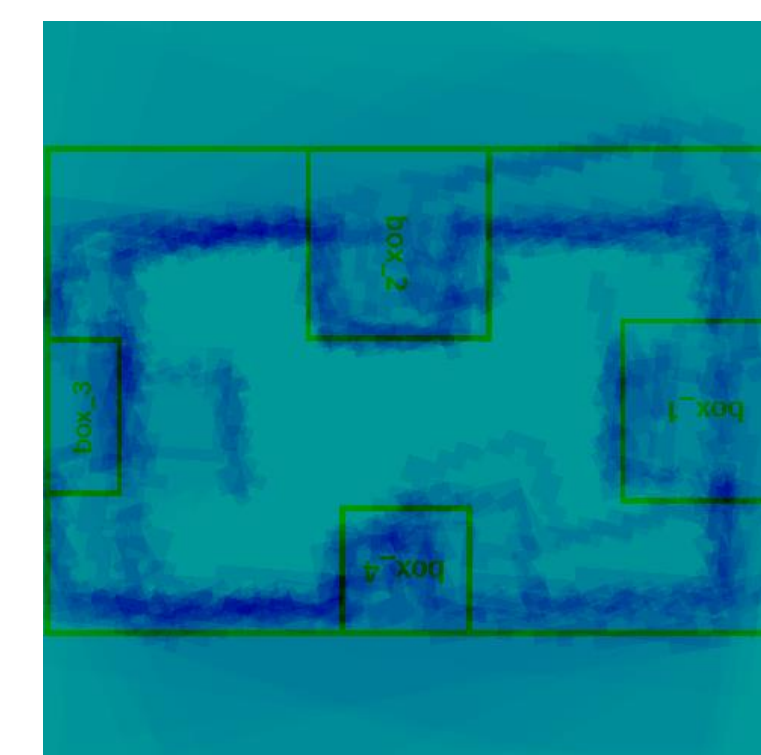


SLAM Toolbox

- All saved maps were resized to a fixed pixel dimension (500 by 500 pixels) to maintain consistent scaling
- Image registration technique was used to overlay maps and visually align them for direct qualitative comparison



Cartographer



SLAM Toolbox

- The program also computed the map error by quantifying pixel-level deviations between the generated SLAM maps
- Following [4], we measured the average Euclidean distance between each map point and its corresponding ground-truth point

$$e_M = \frac{1}{N} \sum_{i=1}^N \sqrt{(x_i - x_i^{GT})^2 + (y_i - y_i^{GT})^2}$$

- In addition to raw error, we also report Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Pearson's correlation coefficient vs. ground truth

RESULTS

	Cartographer		SLAM Toolbox	
	Error (px)	Error (cm)	Error (px)	Error (cm)
Mean	25.227	126.134	19.294	96.468
Variance	5.517	137.918	7.207	180.167
MSE	641.909	16047.729	379.448	9486.196
RMSE	25.336	126.680	19.479	97.397
Correlation	0.246		0.169	

Scale: 5 cm / px

- In general, SLAM Toolbox outperformed Cartographer, but Cartographer yielded smaller variance across multiple runs and higher correlation to the ground truth

DISCUSSION

- TurtleBot3 with ROS2 provide an ideal testbed for evaluating SLAM algorithm accuracy and reliability in real-world robotics
- The lower average map error produced by SLAM Toolbox means the robot's perceived environment was physically closer to the true maze geometry
- SLAM Toolbox's performance suggests stronger handling of local sensor noise and small-scale geometric deviations
- Cartographer may smooth map features more aggressively, reducing noise but also increasing systematic drift in constrained environments like the maze

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